

75 DSA Questions from Leet-Code

1. Arrays (10 Questions)

1. Two Sum
121. Best Time to Buy and Sell Stock
88. Merge Sorted Array
217. Contains Duplicate
238. Product of Array Except Self
53. Maximum Subarray
15. 3Sum
56. Merge Intervals
11. Container With Most Water
48. Rotate Image

Solution:

1. Two Sum (LeetCode 1)

Problem: Find two numbers in the array that add up to a specific target.

Solution:

```
cpp
Copy code
vector<int> twoSum(vector<int>& nums, int target) {
    unordered_map<int, int> map; // Value to index map
    for (int i = 0; i < nums.size(); ++i) {
        int complement = target - nums[i];
        if (map.find(complement) != map.end()) {
            return {map[complement], i};
        }
        map[nums[i]] = i;
    }
    return {};
}
```

2. Best Time to Buy and Sell Stock (LeetCode 121)

Problem: Maximize profit by choosing one day to buy and another to sell.

Solution:

```
cpp
Copy code
int maxProfit(vector<int>& prices) {
    int minPrice = INT_MAX, maxProfit = 0;
    for (int price : prices) {
        minPrice = min(minPrice, price);
        maxProfit = max(maxProfit, price - minPrice);
    }
}
```

```
    }  
    return maxProfit;  
}
```

3. Merge Sorted Array (LeetCode 88)

Problem: Merge two sorted arrays into one sorted array.

Solution:

```
cpp  
Copy code  
void merge(vector<int>& nums1, int m, vector<int>& nums2, int n) {  
    int i = m - 1, j = n - 1, k = m + n - 1;  
    while (i >= 0 && j >= 0) {  
        nums1[k--] = (nums1[i] > nums2[j]) ? nums1[i--] : nums2[j--];  
    }  
    while (j >= 0) {  
        nums1[k--] = nums2[j--];  
    }  
}
```

4. Contains Duplicate (LeetCode 217)

Problem: Check if an array contains duplicates.

Solution:

```
cpp  
Copy code  
bool containsDuplicate(vector<int>& nums) {  
    unordered_set<int> set;  
    for (int num : nums) {  
        if (set.count(num)) return true;  
        set.insert(num);  
    }  
    return false;  
}
```

5. Product of Array Except Self (LeetCode 238)

Problem: Return an array where each element is the product of all other elements.

Solution:

```
cpp  
Copy code  
vector<int> productExceptSelf(vector<int>& nums) {  
    int n = nums.size();  
    vector<int> res(n, 1);  
    int left = 1, right = 1;  
    for (int i = 0; i < n; ++i) {  
        res[i] *= left;  
        left *= nums[i];  
        res[n - 1 - i] *= right;  
        right *= nums[n - 1 - i];  
    }  
    return res;  
}
```

6. Maximum Subarray (LeetCode 53)

Problem: Find the contiguous subarray with the largest sum.

Solution:

```
cpp
Copy code
int maxSubArray(vector<int>& nums) {
    int maxSum = nums[0], currentSum = nums[0];
    for (int i = 1; i < nums.size(); ++i) {
        currentSum = max(nums[i], currentSum + nums[i]);
        maxSum = max(maxSum, currentSum);
    }
    return maxSum;
}
```

7. 3Sum (LeetCode 15)

Problem: Find all unique triplets in the array which gives the sum of zero.

Solution:

```
cpp
Copy code
vector<vector<int>> threeSum(vector<int>& nums) {
    sort(nums.begin(), nums.end());
    vector<vector<int>> res;
    for (int i = 0; i < nums.size() - 2; ++i) {
        if (i > 0 && nums[i] == nums[i - 1]) continue;
        int left = i + 1, right = nums.size() - 1;
        while (left < right) {
            int sum = nums[i] + nums[left] + nums[right];
            if (sum == 0) {
                res.push_back({nums[i], nums[left], nums[right]});
                while (left < right && nums[left] == nums[left + 1]) ++left;
                while (left < right && nums[right] == nums[right - 1])
                    --right;
                ++left; --right;
            } else if (sum < 0) {
                ++left;
            } else {
                --right;
            }
        }
    }
    return res;
}
```

8. Merge Intervals (LeetCode 56)

Problem: Merge overlapping intervals.

Solution:

```
cpp
Copy code
vector<vector<int>> merge(vector<vector<int>>& intervals) {
    sort(intervals.begin(), intervals.end());
    vector<vector<int>> merged;
    for (auto& interval : intervals) {
        if (merged.empty() || merged.back()[1] < interval[0]) {
```

```

        merged.push_back(interval);
    } else {
        merged.back()[1] = max(merged.back()[1], interval[1]);
    }
}
return merged;
}

```

9. Container With Most Water (LeetCode 11)

Problem: Find the maximum water that can be trapped between two lines.

Solution:

```

cpp
Copy code
int maxArea(vector<int>& height) {
    int left = 0, right = height.size() - 1, maxWater = 0;
    while (left < right) {
        maxWater = max(maxWater, min(height[left], height[right]) * (right -
left));
        if (height[left] < height[right]) ++left;
        else --right;
    }
    return maxWater;
}

```

10. Rotate Image (LeetCode 48)

Problem: Rotate a matrix 90 degrees clockwise.

Solution:

```

cpp
Copy code
void rotate(vector<vector<int>>& matrix) {
    int n = matrix.size();
    for (int i = 0; i < n; ++i) {
        for (int j = i; j < n; ++j) {
            swap(matrix[i][j], matrix[j][i]);
        }
    }
    for (auto& row : matrix) {
        reverse(row.begin(), row.end());
    }
}

```

2. Strings (10 Questions)

1. **20. Valid Parentheses**
2. **125. Valid Palindrome**
3. **242. Valid Anagram**
4. **49. Group Anagrams**
5. **5. Longest Palindromic Substring**
6. **76. Minimum Window Substring**
7. **28. Find the Index of the First Occurrence in a String**
8. **443. String Compression**
9. **14. Longest Common Prefix**

10. 459. Repeated Substring Pattern

Solution:

1. Valid Parentheses (LeetCode 20)

```
cpp
Copy code
bool isValid(string s) {
    stack<char> st;
    unordered_map<char, char> map = {{'}','('}, {'}','{'}, {'}','['}, {'}','('}, {'(','}'}, {'(','['}, {'(','('}};
    for (char c : s) {
        if (map.count(c)) {
            if (st.empty() || st.top() != map[c]) return false;
            st.pop();
        } else {
            st.push(c);
        }
    }
    return st.empty();
}
```

2. Valid Palindrome (LeetCode 125)

```
cpp
Copy code
bool isPalindrome(string s) {
    int left = 0, right = s.size() - 1;
    while (left < right) {
        while (left < right && !isalnum(s[left])) ++left;
        while (left < right && !isalnum(s[right])) --right;
        if (tolower(s[left]) != tolower(s[right])) return false;
        ++left; --right;
    }
    return true;
}
```

3. Valid Anagram (LeetCode 242)

```
cpp
Copy code
bool isAnagram(string s, string t) {
    if (s.size() != t.size()) return false;
    vector<int> count(26, 0);
    for (char c : s) count[c - 'a']++;
    for (char c : t) {
        if (--count[c - 'a'] < 0) return false;
    }
    return true;
}
```

4. Group Anagrams (LeetCode 49)

```
cpp
Copy code
vector<vector<string>> groupAnagrams(vector<string>& strs) {
    unordered_map<string, vector<string>> map;
```

```

    for (string s : strs) {
        string sorted = s;
        sort(sorted.begin(), sorted.end());
        map[sorted].push_back(s);
    }
    vector<vector<string>> result;
    for (auto& pair : map) result.push_back(pair.second);
    return result;
}

```

5. Longest Palindromic Substring (LeetCode 5)

cpp

Copy code

```

string longestPalindrome(string s) {
    int n = s.size(), start = 0, maxLength = 1;
    vector<vector<bool>> dp(n, vector<bool>(n, false));
    for (int i = 0; i < n; ++i) dp[i][i] = true;
    for (int i = n - 1; i >= 0; --i) {
        for (int j = i + 1; j < n; ++j) {
            if (s[i] == s[j] && (j - i == 1 || dp[i + 1][j - 1])) {
                dp[i][j] = true;
                if (j - i + 1 > maxLength) {
                    start = i;
                    maxLength = j - i + 1;
                }
            }
        }
    }
    return s.substr(start, maxLength);
}

```

6. Minimum Window Substring (LeetCode 76)

cpp

Copy code

```

string minWindow(string s, string t) {
    unordered_map<char, int> freq;
    for (char c : t) freq[c]++;
    int left = 0, minLen = INT_MAX, count = 0, start = 0;
    for (int right = 0; right < s.size(); ++right) {
        if (--freq[s[right]] >= 0) ++count;
        while (count == t.size()) {
            if (minLen > right - left + 1) {
                minLen = right - left + 1;
                start = left;
            }
            if (++freq[s[left++]] > 0) --count;
        }
    }
    return minLen == INT_MAX ? "" : s.substr(start, minLen);
}

```

7. Find the Index of the First Occurrence (LeetCode 28)

cpp

Copy code

```

int strStr(string haystack, string needle) {
    int m = haystack.size(), n = needle.size();

```

```

    for (int i = 0; i <= m - n; ++i) {
        if (haystack.substr(i, n) == needle) return i;
    }
    return -1;
}

```

8. String Compression (LeetCode 443)

```

cpp
Copy code
int compress(vector<char>& chars) {
    int index = 0, n = chars.size();
    for (int i = 0; i < n; i++) {
        char c = chars[i];
        int count = 1;
        while (i + 1 < n && chars[i + 1] == c) {
            ++i;
            ++count;
        }
        chars[index++] = c;
        if (count > 1) {
            for (char digit : to_string(count)) chars[index++] = digit;
        }
    }
    return index;
}

```

9. Longest Common Prefix (LeetCode 14)

```

cpp
Copy code
string longestCommonPrefix(vector<string>& strs) {
    if (strs.empty()) return "";
    string prefix = strs[0];
    for (int i = 1; i < strs.size(); ++i) {
        while (strs[i].find(prefix) != 0) {
            prefix = prefix.substr(0, prefix.size() - 1);
            if (prefix.empty()) return "";
        }
    }
    return prefix;
}

```

10. Repeated Substring Pattern (LeetCode 459)

```

cpp
Copy code
bool repeatedSubstringPattern(string s) {
    string doubled = s + s;
    return doubled.substr(1, doubled.size() - 2).find(s) != string::npos;
}

```

Linked Lists (8 Questions)

1. **206. Reverse Linked List**
2. **21. Merge Two Sorted Lists**
3. **19. Remove Nth Node From End of List**
4. **141. Linked List Cycle**
5. **2. Add Two Numbers**
6. **160. Intersection of Two Linked Lists**
7. **234. Palindrome Linked List**
8. **25. Reverse Nodes in k-Group**

Solution:

Linked Lists

1. Reverse Linked List (LeetCode 206)

```
cpp
Copy code
ListNode* reverseList(ListNode* head) {
    ListNode* prev = nullptr;
    while (head) {
        ListNode* nextNode = head->next;
        head->next = prev;
        prev = head;
        head = nextNode;
    }
    return prev;
}
```

2. Merge Two Sorted Lists (LeetCode 21)

```
cpp
Copy code
ListNode* mergeTwoLists(ListNode* list1, ListNode* list2) {
    if (!list1) return list2;
    if (!list2) return list1;
    if (list1->val < list2->val) {
        list1->next = mergeTwoLists(list1->next, list2);
        return list1;
    } else {
        list2->next = mergeTwoLists(list1, list2->next);
        return list2;
    }
}
```

3. Remove Nth Node From End of List (LeetCode 19)

```
cpp
Copy code
ListNode* removeNthFromEnd(ListNode* head, int n) {
    ListNode dummy(0, head);
    ListNode* slow = &dummy, *fast = &dummy;
    for (int i = 0; i <= n; ++i) fast = fast->next;
    while (fast) {
        slow = slow->next;
        fast = fast->next;
    }
    slow->next = slow->next->next;
}
```



```
        return dummy.next;
    }
}
```

4. Linked List Cycle (LeetCode 141)

```
cpp
Copy code
bool hasCycle(ListNode* head) {
    ListNode* slow = head, *fast = head;
    while (fast && fast->next) {
        slow = slow->next;
        fast = fast->next->next;
        if (slow == fast) return true;
    }
    return false;
}
```

5. Add Two Numbers (LeetCode 2)

```
cpp
Copy code
ListNode* addTwoNumbers(ListNode* l1, ListNode* l2) {
    ListNode dummy, *current = &dummy;
    int carry = 0;
    while (l1 || l2 || carry) {
        int sum = carry;
        if (l1) sum += l1->val, l1 = l1->next;
        if (l2) sum += l2->val, l2 = l2->next;
        carry = sum / 10;
        current->next = new ListNode(sum % 10);
        current = current->next;
    }
    return dummy.next;
}
```

6. Intersection of Two Linked Lists (LeetCode 160)

```
cpp
Copy code
ListNode* getIntersectionNode(ListNode* headA, ListNode* headB) {
    ListNode* a = headA, *b = headB;
    while (a != b) {
        a = a ? a->next : headB;
        b = b ? b->next : headA;
    }
    return a;
}
```

7. Palindrome Linked List (LeetCode 234)

```
cpp
Copy code
bool isPalindrome(ListNode* head) {
    ListNode* slow = head, *fast = head, *prev = nullptr;
    while (fast && fast->next) {
        fast = fast->next->next;
        ListNode* temp = slow->next;
        slow->next = prev;
    }
```

```

        prev = slow;
        slow = temp;
    }
    if (fast) slow = slow->next;
    while (slow) {
        if (slow->val != prev->val) return false;
        slow = slow->next;
        prev = prev->next;
    }
    return true;
}

```

8. Reverse Nodes in k-Group (LeetCode 25)

cpp

Copy code

```

ListNode* reverseKGroup(ListNode* head, int k) {
    ListNode* temp = head;
    for (int i = 0; i < k; ++i) {
        if (!temp) return head;
        temp = temp->next;
    }
    ListNode* prev = nullptr, *current = head, *next = nullptr;
    for (int i = 0; i < k; ++i) {
        next = current->next;
        current->next = prev;
        prev = current;
        current = next;
    }
    head->next = reverseKGroup(next, k);
    return prev;
}

```

4. Stacks and Queues (6 Questions)

1. **232. Implement Queue using Stacks**
2. **225. Implement Stack using Queues**
3. **739. Daily Temperatures**
4. **496. Next Greater Element I**
5. **20. Valid Parentheses**
6. **155. Min Stack**

Solution:

[Stacks and Queues](#)

1. Implement Queue Using Stacks (LeetCode 232)

cpp

Copy code

```
class MyQueue {
    stack<int> s1, s2;
public:
    void push(int x) {
        s1.push(x);
    }
    int pop() {
        peek();
        int top = s2.top();
        s2.pop();
        return top;
    }
    int peek() {
        if (s2.empty()) {
            while (!s1.empty()) {
                s2.push(s1.top());
                s1.pop();
            }
        }
        return s2.top();
    }
    bool empty() {
        return s1.empty() && s2.empty();
    }
};
```

2. Implement Stack Using Queues (LeetCode 225)

cpp

Copy code

```
class MyStack {
    queue<int> q1, q2;
public:
    void push(int x) {
        q2.push(x);
        while (!q1.empty()) {
            q2.push(q1.front());
            q1.pop();
        }
        swap(q1, q2);
    }
    int pop() {
        int top = q1.front();
        q1.pop();
        return top;
    }
    int top() {
        return q1.front();
    }
    bool empty() {
        return q1.empty();
    }
};
```

3. Min Stack (LeetCode 155)

```

cpp
Copy code
class MinStack {
    stack<int> s, minS;
public:
    void push(int val) {
        s.push(val);
        if (minS.empty() || val <= minS.top()) minS.push(val);
    }
    void pop() {
        if (s.top() == minS.top()) minS.pop();
        s.pop();
    }
    int top() {
        return s.top();
    }
    int getMin() {
        return minS.top();
    }
};

```

4. Daily Temperatures (LeetCode 739)

```

cpp
Copy code
vector<int> dailyTemperatures(vector<int>& temperatures) {
    stack<int> s;
    vector<int> res(temperatures.size(), 0);
    for (int i = 0; i < temperatures.size(); ++i) {
        while (!s.empty() && temperatures[i] > temperatures[s.top()]) {
            int index = s.top();
            s.pop();
            res[index] = i - index;
        }
        s.push(i);
    }
    return res;
}

```

5. Evaluate Reverse Polish Notation (LeetCode 150)

```

cpp
Copy code
int evalRPN(vector<string>& tokens) {
    stack<int> s;
    for (string& token : tokens) {
        if (token == "+" || token == "-" || token == "*" || token == "/") {
            int b = s.top(); s.pop();
            int a = s.top(); s.pop();
            if (token == "+") s.push(a + b);
            else if (token == "-") s.push(a - b);
            else if (token == "*") s.push(a * b);
            else s.push(a / b);
        } else {
            s.push(stoi(token));
        }
    }
    return s.top();
}

```

6. Next Greater Element I (LeetCode 496)

cpp

Copy code

```
vector<int> nextGreaterElement(vector<int>& nums1, vector<int>& nums2) {
    unordered_map<int, int> map;
    stack<int> s;
    for (int num : nums2) {
        while (!s.empty() && s.top() < num) {
            map[s.top()] = num;
            s.pop();
        }
        s.push(num);
    }
    vector<int> res;
    for (int num : nums1) {
        res.push_back(map.count(num) ? map[num] : -1);
    }
    return res;
}
```

7. Next Greater Element II (LeetCode 503)

cpp

Copy code

```
vector<int> nextGreaterElements(vector<int>& nums) {
    int n = nums.size();
    vector<int> res(n, -1);
    stack<int> s;
    for (int i = 0; i < 2 * n; ++i) {
        while (!s.empty() && nums[s.top()] < nums[i % n]) {
            res[s.top()] = nums[i % n];
            s.pop();
        }
        if (i < n) s.push(i);
    }
    return res;
}
```

8. Circular Queue (LeetCode 622)

cpp

Copy code

```
class MyCircularQueue {
    vector<int> q;
    int head, tail, size;
public:
    MyCircularQueue(int k) : q(k), head(-1), tail(-1), size(k) {}
    bool enqueue(int value) {
        if (isFull()) return false;
        if (isEmpty()) head = 0;
        tail = (tail + 1) % size;
        q[tail] = value;
        return true;
    }
    bool dequeue() {
        if (isEmpty()) return false;
        if (head == tail) head = tail = -1;
    }
};
```

```

        else head = (head + 1) % size;
        return true;
    }
    int Front() {
        return isEmpty() ? -1 : q[head];
    }
    int Rear() {
        return isEmpty() ? -1 : q[tail];
    }
    bool isEmpty() {
        return head == -1;
    }
    bool isFull() {
        return (tail + 1) % size == head;
    }
};

```

5. Binary Search (6 Questions)

1. **704. Binary Search**
2. **34. Find First and Last Position of Element in Sorted Array**
3. **74. Search a 2D Matrix**
4. **33. Search in Rotated Sorted Array**
5. **81. Search in Rotated Sorted Array II**
6. **162. Find Peak Element**

Solution:

704. Binary Search

```

cpp
Copy code
int search(vector<int>& nums, int target) {
    int left = 0, right = nums.size() - 1;
    while (left <= right) {
        int mid = left + (right - left) / 2;
        if (nums[mid] == target) return mid;
        if (nums[mid] < target) left = mid + 1;
        else right = mid - 1;
    }
    return -1;
}

```

34. Find First and Last Position of Element in Sorted Array

```

cpp
Copy code
vector<int> searchRange(vector<int>& nums, int target) {

```

```

int left = 0, right = nums.size() - 1;
vector<int> result(2, -1);

// Find the first position
while (left <= right) {
    int mid = left + (right - left) / 2;
    if (nums[mid] < target) left = mid + 1;
    else right = mid - 1;
    if (nums[mid] == target) result[0] = mid;
}

left = 0, right = nums.size() - 1;
// Find the last position
while (left <= right) {
    int mid = left + (right - left) / 2;
    if (nums[mid] <= target) left = mid + 1;
    else right = mid - 1;
    if (nums[mid] == target) result[1] = mid;
}

return result;
}

```

74. Search a 2D Matrix

cpp

Copy code

```

bool searchMatrix(vector<vector<int>>& matrix, int target) {
    int rows = matrix.size(), cols = matrix[0].size();
    int left = 0, right = rows * cols - 1;
    while (left <= right) {
        int mid = left + (right - left) / 2;
        int midElement = matrix[mid / cols][mid % cols];
        if (midElement == target) return true;
        if (midElement < target) left = mid + 1;
        else right = mid - 1;
    }
    return false;
}

```

33. Search in Rotated Sorted Array

cpp

Copy code

```

int search(vector<int>& nums, int target) {
    int left = 0, right = nums.size() - 1;
    while (left <= right) {
        int mid = left + (right - left) / 2;
        if (nums[mid] == target) return mid;

        // Check which side is sorted
        if (nums[left] <= nums[mid]) { // Left side is sorted
            if (nums[left] <= target && target < nums[mid]) right = mid - 1;
            else left = mid + 1;
        } else { // Right side is sorted
            if (nums[mid] < target && target <= nums[right]) left = mid + 1;
            else right = mid - 1;
        }
    }
}

```

```
        return -1;
    }
}
```

81. Search in Rotated Sorted Array II

cpp

Copy code

```
bool search(vector<int>& nums, int target) {
    int left = 0, right = nums.size() - 1;
    while (left <= right) {
        int mid = left + (right - left) / 2;
        if (nums[mid] == target) return true;

        // Handle duplicates
        if (nums[left] == nums[mid] && nums[right] == nums[mid]) {
            left++;
            right--;
        } else if (nums[left] <= nums[mid]) { // Left side is sorted
            if (nums[left] <= target && target < nums[mid]) right = mid - 1;
            else left = mid + 1;
        } else { // Right side is sorted
            if (nums[mid] < target && target <= nums[right]) left = mid + 1;
            else right = mid - 1;
        }
    }
    return false;
}
```

162. Find Peak Element

cpp

Copy code

```
int findPeakElement(vector<int>& nums) {
    int left = 0, right = nums.size() - 1;
    while (left < right) {
        int mid = left + (right - left) / 2;
        if (nums[mid] > nums[mid + 1]) right = mid;
        else left = mid + 1;
    }
    return left;
}
```

6. Trees (8 Questions)

1. **104. Maximum Depth of Binary Tree**
2. **100. Same Tree**
3. **101. Symmetric Tree**
4. **144. Binary Tree Preorder Traversal**
5. **94. Binary Tree Inorder Traversal**
6. **145. Binary Tree Postorder Traversal**
7. **102. Binary Tree Level Order Traversal**
8. **110. Balanced Binary Tree**

Solution:

104. Maximum Depth of Binary Tree

```
cpp
Copy code
int maxDepth(TreeNode* root) {
    if (!root) return 0;
    return 1 + max(maxDepth(root->left), maxDepth(root->right));
}
```

100. Same Tree

```
cpp
Copy code
bool isSameTree(TreeNode* p, TreeNode* q) {
    if (!p && !q) return true;
    if (!p || !q || p->val != q->val) return false;
    return isSameTree(p->left, q->left) && isSameTree(p->right, q->right);
}
```

101. Symmetric Tree

```
cpp
Copy code
bool isMirror(TreeNode* t1, TreeNode* t2) {
    if (!t1 && !t2) return true;
    if (!t1 || !t2 || t1->val != t2->val) return false;
    return isMirror(t1->left, t2->right) && isMirror(t1->right, t2->left);
}

bool isSymmetric(TreeNode* root) {
    return isMirror(root, root);
}
```

144. Binary Tree Preorder Traversal

```
cpp
Copy code
vector<int> preorderTraversal(TreeNode* root) {
    vector<int> result;
    stack<TreeNode*> st;
    if (root) st.push(root);
    while (!st.empty()) {
        TreeNode* curr = st.top();
        st.pop();
        result.push_back(curr->val);
        if (curr->right) st.push(curr->right);
        if (curr->left) st.push(curr->left);
    }
    return result;
}
```

94. Binary Tree Inorder Traversal

cpp

Copy code

```
vector<int> inorderTraversal(TreeNode* root) {
    vector<int> result;
    stack<TreeNode*> st;
    TreeNode* curr = root;
    while (curr || !st.empty()) {
        while (curr) {
            st.push(curr);
            curr = curr->left;
        }
        curr = st.top();
        st.pop();
        result.push_back(curr->val);
        curr = curr->right;
    }
    return result;
}
```

145. Binary Tree Postorder Traversal

cpp

Copy code

```
vector<int> postorderTraversal(TreeNode* root) {
    vector<int> result;
    stack<TreeNode*> st;
    TreeNode* lastVisited = nullptr;
    while (root || !st.empty()) {
        if (root) {
            st.push(root);
            root = root->left;
        } else {
            TreeNode* node = st.top();
            if (node->right && node->right != lastVisited) {
                root = node->right;
            } else {
                result.push_back(node->val);
                lastVisited = node;
                st.pop();
            }
        }
    }
    return result;
}
```

102. Binary Tree Level Order Traversal

cpp

Copy code

```
vector<vector<int>>> levelOrder(TreeNode* root) {
    vector<vector<int>>> result;
    if (!root) return result;
    queue<TreeNode*> q;
    q.push(root);
    while (!q.empty()) {
        int size = q.size();
        vector<int> level;
```

```

        for (int i = 0; i < size; ++i) {
            TreeNode* curr = q.front();
            q.pop();
            level.push_back(curr->val);
            if (curr->left) q.push(curr->left);
            if (curr->right) q.push(curr->right);
        }
        result.push_back(level);
    }
    return result;
}

```

110. Balanced Binary Tree

cpp

Copy code

```

int height(TreeNode* root) {
    if (!root) return 0;
    int leftHeight = height(root->left);
    int rightHeight = height(root->right);
    if (leftHeight == -1 || rightHeight == -1 || abs(leftHeight -
rightHeight) > 1)
        return -1;
    return 1 + max(leftHeight, rightHeight);
}

bool isBalanced(TreeNode* root) {
    return height(root) != -1;
}

```

7. Recursion and Backtracking (7 Questions)

1. **39. Combination Sum**
2. **46. Permutations**
3. **78. Subsets**
4. **51. N-Queens**
5. **17. Letter Combinations of a Phone Number**
6. **90. Subsets II**
7. **37. Sudoku Solver**

Solution:

39. Combination Sum

cpp

Copy code

```

void backtrack(vector<int>& candidates, int target, int start, vector<int>&
current, vector<vector<int>>& result) {
    if (target == 0) {
        result.push_back(current);
        return;
    }
    for (int i = start; i < candidates.size(); i++) {
        if (candidates[i] > target) continue;
        current.push_back(candidates[i]);
    }
}

```

```

        backtrack(candidates, target - candidates[i], i, current, result);
        current.pop_back();
    }
}

vector<vector<int>> combinationSum(vector<int>& candidates, int target) {
    vector<vector<int>> result;
    vector<int> current;
    backtrack(candidates, target, 0, current, result);
    return result;
}

```

46. Permutations

cpp

Copy code

```

void permuteHelper(vector<int>& nums, int start, vector<vector<int>>& result)
{
    if (start == nums.size()) {
        result.push_back(nums);
        return;
    }
    for (int i = start; i < nums.size(); i++) {
        swap(nums[start], nums[i]);
        permuteHelper(nums, start + 1, result);
        swap(nums[start], nums[i]);
    }
}

vector<vector<int>> permute(vector<int>& nums) {
    vector<vector<int>> result;
    permuteHelper(nums, 0, result);
    return result;
}

```

78. Subsets

cpp

Copy code

```

void backtrack(vector<int>& nums, int start, vector<int>& current,
vector<vector<int>>& result) {
    result.push_back(current);
    for (int i = start; i < nums.size(); i++) {
        current.push_back(nums[i]);
        backtrack(nums, i + 1, current, result);
        current.pop_back();
    }
}

vector<vector<int>> subsets(vector<int>& nums) {
    vector<vector<int>> result;
    vector<int> current;
    backtrack(nums, 0, current, result);
    return result;
}

```

51. N-Queens

cpp

Copy code

```
bool isSafe(vector<string>& board, int row, int col, int n) {
    for (int i = 0; i < row; i++)
        if (board[i][col] == 'Q') return false;
    for (int i = row - 1, j = col - 1; i >= 0 && j >= 0; i--, j--)
        if (board[i][j] == 'Q') return false;
    for (int i = row - 1, j = col + 1; i >= 0 && j < n; i--, j++)
        if (board[i][j] == 'Q') return false;
    return true;
}

void solve(int row, vector<string>& board, vector<vector<string>>& result,
int n) {
    if (row == n) {
        result.push_back(board);
        return;
    }
    for (int col = 0; col < n; col++) {
        if (isSafe(board, row, col, n)) {
            board[row][col] = 'Q';
            solve(row + 1, board, result, n);
            board[row][col] = '.';
        }
    }
}

vector<vector<string>> solveNQueens(int n) {
    vector<vector<string>> result;
    vector<string> board(n, string(n, '.'));
    solve(0, board, result, n);
    return result;
}
```

17. Letter Combinations of a Phone Number

cpp

Copy code

```
void backtrack(string& digits, int index, string& current, vector<string>&
result, vector<string>& mapping) {
    if (index == digits.size()) {
        result.push_back(current);
        return;
    }
    for (char ch : mapping[digits[index] - '0']) {
        current.push_back(ch);
        backtrack(digits, index + 1, current, result, mapping);
        current.pop_back();
    }
}

vector<string> letterCombinations(string digits) {
    if (digits.empty()) return {};
    vector<string> mapping = {"", "", "abc", "def", "ghi", "jkl", "mno",
"pqrs", "tuv", "wxyz"};
    vector<string> result;
    string current;
    backtrack(digits, 0, current, result, mapping);
    return result;
}
```

90. Subsets II

cpp

Copy code

```
void backtrack(vector<int>& nums, int start, vector<int>& current,
vector<vector<int>>& result) {
    result.push_back(current);
    for (int i = start; i < nums.size(); i++) {
        if (i > start && nums[i] == nums[i - 1]) continue;
        current.push_back(nums[i]);
        backtrack(nums, i + 1, current, result);
        current.pop_back();
    }
}

vector<vector<int>> subsetsWithDup(vector<int>& nums) {
    sort(nums.begin(), nums.end());
    vector<vector<int>> result;
    vector<int> current;
    backtrack(nums, 0, current, result);
    return result;
}
```

37. Sudoku Solver

cpp

Copy code

```
bool isValid(vector<vector<char>>& board, int row, int col, char c) {
    for (int i = 0; i < 9; i++) {
        if (board[i][col] == c || board[row][i] == c || board[row / 3 * 3 + i
/ 3][col / 3 * 3 + i % 3] == c)
            return false;
    }
    return true;
}

bool solve(vector<vector<char>>& board) {
    for (int i = 0; i < 9; i++) {
        for (int j = 0; j < 9; j++) {
            if (board[i][j] == '.') {
                for (char c = '1'; c <= '9'; c++) {
                    if (isValid(board, i, j, c)) {
                        board[i][j] = c;
                        if (solve(board)) return true;
                        board[i][j] = '.';
                    }
                }
                return false;
            }
        }
    }
    return true;
}

void solveSudoku(vector<vector<char>>& board) {
    solve(board);
}
```

8. Dynamic Programming (10 Questions)

1. **70. Climbing Stairs**
2. **198. House Robber**
3. **322. Coin Change**
4. **300. Longest Increasing Subsequence**
5. **1143. Longest Common Subsequence**
6. **62. Unique Paths**
7. **5. Longest Palindromic Substring**
8. **718. Maximum Length of Repeated Subarray**
9. **416. Partition Equal Subset Sum**
10. **53. Maximum Subarray**

Solution:

70. Climbing Stairs

```
cpp
Copy code
int climbStairs(int n) {
    if (n <= 2) return n;
    int a = 1, b = 2;
    for (int i = 3; i <= n; i++) {
        int c = a + b;
        a = b;
        b = c;
    }
    return b;
}
```

198. House Robber

```
cpp
Copy code
int rob(vector<int>& nums) {
    int prev1 = 0, prev2 = 0;
    for (int num : nums) {
        int temp = max(prev1, prev2 + num);
        prev2 = prev1;
        prev1 = temp;
    }
    return prev1;
}
```

322. Coin Change

```
cpp
Copy code
int coinChange(vector<int>& coins, int amount) {
    vector<int> dp(amount + 1, amount + 1);
    dp[0] = 0;
    for (int coin : coins) {
        for (int j = coin; j <= amount; j++) {
            dp[j] = min(dp[j], dp[j - coin] + 1);
        }
    }
}
```

```

    }
    return dp[amount] > amount ? -1 : dp[amount];
}

```

300. Longest Increasing Subsequence

```

cpp
Copy code
int lengthOfLIS(vector<int>& nums) {
    vector<int> dp(nums.size(), 1);
    int maxLength = 1;
    for (int i = 1; i < nums.size(); i++) {
        for (int j = 0; j < i; j++) {
            if (nums[i] > nums[j]) {
                dp[i] = max(dp[i], dp[j] + 1);
            }
        }
        maxLength = max(maxLength, dp[i]);
    }
    return maxLength;
}

```

1143. Longest Common Subsequence

```

cpp
Copy code
int longestCommonSubsequence(string text1, string text2) {
    int m = text1.size(), n = text2.size();
    vector<vector<int>> dp(m + 1, vector<int>(n + 1, 0));
    for (int i = 1; i <= m; i++) {
        for (int j = 1; j <= n; j++) {
            if (text1[i - 1] == text2[j - 1])
                dp[i][j] = dp[i - 1][j - 1] + 1;
            else
                dp[i][j] = max(dp[i - 1][j], dp[i][j - 1]);
        }
    }
    return dp[m][n];
}

```

62. Unique Paths

```

cpp
Copy code
int uniquePaths(int m, int n) {
    vector<vector<int>> dp(m, vector<int>(n, 1));
    for (int i = 1; i < m; i++) {
        for (int j = 1; j < n; j++) {
            dp[i][j] = dp[i - 1][j] + dp[i][j - 1];
        }
    }
    return dp[m - 1][n - 1];
}

```

5. Longest Palindromic Substring

```

cpp
Copy code

```



```

string longestPalindrome(string s) {
    int n = s.size();
    vector<vector<bool>> dp(n, vector<bool>(n, false));
    int maxLength = 1, start = 0;

    for (int i = 0; i < n; i++) dp[i][i] = true;

    for (int len = 2; len <= n; len++) {
        for (int i = 0; i <= n - len; i++) {
            int j = i + len - 1;
            if (s[i] == s[j] && (len == 2 || dp[i + 1][j - 1])) {
                dp[i][j] = true;
                if (len > maxLength) {
                    maxLength = len;
                    start = i;
                }
            }
        }
    }

    return s.substr(start, maxLength);
}

```

718. Maximum Length of Repeated Subarray

cpp

Copy code

```

int findLength(vector<int>& nums1, vector<int>& nums2) {
    int m = nums1.size(), n = nums2.size();
    vector<vector<int>> dp(m + 1, vector<int>(n + 1, 0));
    int maxLength = 0;
    for (int i = 1; i <= m; i++) {
        for (int j = 1; j <= n; j++) {
            if (nums1[i - 1] == nums2[j - 1]) {
                dp[i][j] = dp[i - 1][j - 1] + 1;
                maxLength = max(maxLength, dp[i][j]);
            }
        }
    }
    return maxLength;
}

```

416. Partition Equal Subset Sum

cpp

Copy code

```

bool canPartition(vector<int>& nums) {
    int sum = accumulate(nums.begin(), nums.end(), 0);
    if (sum % 2 != 0) return false;
    int target = sum / 2;
    vector<bool> dp(target + 1, false);
    dp[0] = true;

    for (int num : nums) {
        for (int j = target; j >= num; j--) {
            dp[j] = dp[j] || dp[j - num];
        }
    }
    return dp[target];
}

```

53. Maximum Subarray

cpp

Copy code

```
int maxSubArray(vector<int>& nums) {
    int maxSum = nums[0], currentSum = nums[0];
    for (int i = 1; i < nums.size(); i++) {
        currentSum = max(nums[i], currentSum + nums[i]);
        maxSum = max(maxSum, currentSum);
    }
    return maxSum;
}
```

9. Graphs (6 Questions)

1. **133. Clone Graph**
2. **200. Number of Islands**
3. **207. Course Schedule**
4. **785. Is Graph Bipartite?**
5. **994. Rotting Oranges**
6. **323. Number of Connected Components in an Undirected Graph**

Solution:

133. Clone Graph

cpp

Copy code

```
Node* cloneGraph(Node* node) {
    if (!node) return nullptr;

    unordered_map<Node*, Node*> visited;
    queue<Node*> q;
    q.push(node);

    visited[node] = new Node(node->val);

    while (!q.empty()) {
        Node* curr = q.front();
        q.pop();

        for (Node* neighbor : curr->neighbors) {
            if (!visited.count(neighbor)) {
                visited[neighbor] = new Node(neighbor->val);
                q.push(neighbor);
            }
            visited[curr]->neighbors.push_back(visited[neighbor]);
        }
    }
    return visited[node];
}
```

200. Number of Islands

cpp

Copy code

```
void dfs(vector<vector<char>>& grid, int i, int j) {
    if (i < 0 || j < 0 || i >= grid.size() || j >= grid[0].size() ||
        grid[i][j] == '0') return;
    grid[i][j] = '0';
    dfs(grid, i + 1, j);
    dfs(grid, i - 1, j);
    dfs(grid, i, j + 1);
    dfs(grid, i, j - 1);
}

int numIslands(vector<vector<char>>& grid) {
    int count = 0;
    for (int i = 0; i < grid.size(); i++) {
        for (int j = 0; j < grid[0].size(); j++) {
            if (grid[i][j] == '1') {
                count++;
                dfs(grid, i, j);
            }
        }
    }
    return count;
}
```

207. Course Schedule

cpp

Copy code

```
bool canFinish(int numCourses, vector<vector<int>>& prerequisites) {
    vector<vector<int>> graph(numCourses);
    vector<int> inDegree(numCourses, 0);

    for (auto& pre : prerequisites) {
        graph[pre[1]].push_back(pre[0]);
        inDegree[pre[0]]++;
    }

    queue<int> q;
    for (int i = 0; i < numCourses; i++) {
        if (inDegree[i] == 0) q.push(i);
    }

    int count = 0;
    while (!q.empty()) {
        int course = q.front();
        q.pop();
        count++;
        for (int neighbor : graph[course]) {
            if (--inDegree[neighbor] == 0) q.push(neighbor);
        }
    }
    return count == numCourses;
}
```

785. Is Graph Bipartite?

cpp

Copy code

```
bool isBipartite(vector<vector<int>>& graph) {
    int n = graph.size();
    vector<int> color(n, -1);

    for (int i = 0; i < n; i++) {
        if (color[i] != -1) continue;

        queue<int> q;
        q.push(i);
        color[i] = 0;

        while (!q.empty()) {
            int node = q.front();
            q.pop();
            for (int neighbor : graph[node]) {
                if (color[neighbor] == -1) {
                    color[neighbor] = 1 - color[node];
                    q.push(neighbor);
                } else if (color[neighbor] == color[node]) {
                    return false;
                }
            }
        }
    }
    return true;
}
```

994. Rotting Oranges

cpp

Copy code

```
int orangesRotting(vector<vector<int>>& grid) {
    int m = grid.size(), n = grid[0].size();
    queue<pair<int, int>> q;
    int fresh = 0;

    for (int i = 0; i < m; i++) {
        for (int j = 0; j < n; j++) {
            if (grid[i][j] == 2) q.push({i, j});
            if (grid[i][j] == 1) fresh++;
        }
    }

    int minutes = 0;
    vector<pair<int, int>> directions = {{0, 1}, {1, 0}, {0, -1}, {-1, 0}};

    while (!q.empty() && fresh > 0) {
        int size = q.size();
        for (int i = 0; i < size; i++) {
            auto [x, y] = q.front();
            q.pop();
            for (auto [dx, dy] : directions) {
                int nx = x + dx, ny = y + dy;
                if (nx >= 0 && ny >= 0 && nx < m && ny < n && grid[nx][ny] ==
1) {
                    grid[nx][ny] = 2;
                    q.push({nx, ny});
                    fresh--;
                }
            }
        }
    }

    return fresh > 0 ? -1 : minutes;
```

```

        }
    }
    minutes++;
}
return fresh == 0 ? minutes : -1;
}

```

323. Number of Connected Components in an Undirected Graph

cpp

Copy code

```

void dfs(vector<vector<int>>& graph, vector<bool>& visited, int node) {
    visited[node] = true;
    for (int neighbor : graph[node]) {
        if (!visited[neighbor]) {
            dfs(graph, visited, neighbor);
        }
    }
}

int countComponents(int n, vector<vector<int>>& edges) {
    vector<vector<int>> graph(n);
    for (auto& edge : edges) {
        graph[edge[0]].push_back(edge[1]);
        graph[edge[1]].push_back(edge[0]);
    }

    vector<bool> visited(n, false);
    int count = 0;

    for (int i = 0; i < n; i++) {
        if (!visited[i]) {
            count++;
            dfs(graph, visited, i);
        }
    }
    return count;
}

```

10. Bit Manipulation (4 Questions)

1. **136. Single Number**
2. **190. Reverse Bits**
3. **191. Number of 1 Bits**
4. **268. Missing Number**

Solution:

136. Single Number

```

cpp
Copy code
int singleNumber(vector<int>& nums) {
    int result = 0;
    for (int num : nums) {
        result ^= num;
    }
    return result;
}

```

Explanation: XOR operation cancels out numbers that appear twice, leaving only the number that appears once.

190. Reverse Bits

```

cpp
Copy code
uint32_t reverseBits(uint32_t n) {
    uint32_t result = 0;
    for (int i = 0; i < 32; i++) {
        result = (result << 1) | (n & 1);
        n >>= 1;
    }
    return result;
}

```

Explanation: Process each bit of the number, shifting the result left and appending the current bit of `n` to the result.

191. Number of 1 Bits

```

cpp
Copy code
int hammingWeight(uint32_t n) {
    int count = 0;
    while (n != 0) {
        count += n & 1;
        n >>= 1;
    }
    return count;
}

```

Explanation: Count the number of set bits (1s) by repeatedly masking the least significant bit and shifting the number right.

268. Missing Number

```

cpp
Copy code
int missingNumber(vector<int>& nums) {
    int n = nums.size();
    int totalXOR = 0, arrayXOR = 0;
    for (int i = 0; i <= n; i++) {
        totalXOR ^= i;
    }
    for (int num : nums) {
        arrayXOR ^= num;
    }
}

```

```
    }  
    return totalXOR ^ arrayXOR;  
}
```


